Thermal Physics & Thermodynamics

1. Introduction

Thermal physics studies heat, temperature, internal energy, and the laws governing energy transfer in matter. Thermodynamics is the broader framework that describes how energy, work, heat, and entropy behave in physical systems.

For BSc Physics, this subject is central because it connects microscopic molecular motion with macroscopic observables such as pressure, temperature, heat capacity, phase change, and spontaneity.

2. Basic concepts

Thermodynamic system

A thermodynamic system is the part of the universe chosen for study, while the surroundings are everything outside it. Systems may be open, closed, or isolated depending on whether matter and energy can cross the boundary.

State variables

Important state variables include pressure, volume, temperature, internal energy, entropy, enthalpy, and free energy. These depend only on the state of the system, not on the path taken.

Equilibrium

A system is in thermodynamic equilibrium when its macroscopic properties remain unchanged with time and there is no net flow of energy or matter.

3. Temperature and thermal equilibrium

Temperature is the physical quantity that determines the direction of heat flow between bodies. If two systems are each in thermal equilibrium with a third, they are in thermal equilibrium with each other; this is the basis of the zeroth law.

Zeroth law of thermodynamics

If systems A and B are each in thermal equilibrium with system C, then A and B are in thermal equilibrium with each other. This law justifies the concept of temperature.

4. Temperature scales

Temperature may be expressed in Celsius, Fahrenheit, or Kelvin. The Kelvin scale is the absolute thermodynamic temperature scale used in physics.

Key relations

- $T_K = T_{\circ C} + 273.15.$
- $T_{\circ F} = \frac{9}{5} T_{\circ C} + 32.$
- $T_{\circ C} = \frac{5}{9}(T_{\circ F} - 32).$

5. Heat and work

Heat is energy transferred because of temperature difference, while work is energy transferred by generalized forces such as pressure-volume work.

Sign convention

In physics, the first law is commonly written as

$$\Delta U = Q + W$$

where $Q$is heat added to the system and $W$is work done on the system.

Pressure-volume work

For a quasi-static process,

$$dW = -P\, dV$$

and for a finite reversible process,

$$W = -\int_{V_1}^{V_2} P\, dV$$

This is especially important for gases.

6. First law of thermodynamics

The first law is the law of conservation of energy in thermodynamic form. Energy cannot be created or destroyed, only transferred or transformed.

Mathematical form

$$dU = \delta Q + \delta W$$

or for finite changes,

$$\Delta U = Q + W$$

depending on convention.

Meaning

- If heat is supplied, internal energy may increase.
- If the system does work, internal energy decreases unless compensated by heat input.
- In cyclic processes, $\Delta U = 0$

7. Internal energy

Internal energy is the total microscopic energy of a system, including translational, rotational, vibrational, electronic, and interaction energies. It is a state function.

For an ideal gas, internal energy depends only on temperature. This is a key result used in many derivations.

8. Heat capacity and calorimetry

Heat capacity is the heat required to raise the temperature of a body by one degree. Specific heat capacity is heat per unit mass per degree rise in temperature.

Basic formulas

$$Q = C\Delta T$$

$$Q = mc\Delta T$$

where C is heat capacity, m is mass, and c is specific heat.

Molar heat capacity

$$Q = nC_m\Delta T$$

where n is number of moles.

Calorimetry principle

In an isolated system, heat lost by the hotter body equals heat gained by the colder body:

$$Q_{\text{lost}} + Q_{\text{gained}} = 0$$

This is used in mixing and specific heat experiments.

9. Thermal expansion

Most substances expand on heating because atomic vibrations increase with temperature.

Linear expansion

$$\Delta L = \alpha L_0 \Delta T$$

$$L = L_0(1 + \alpha\Delta T)$$

where α is the coefficient of linear expansion.

Area expansion

$$\Delta A = \beta A_0 \Delta T$$

with $\beta \approx 2\alpha$.

Volume expansion

$$\Delta V = \gamma V_0 \Delta T$$

with $\gamma \approx 3\alpha$.

10. Ideal gas and kinetic theory

The ideal gas law links pressure, volume, temperature, and amount of gas.

Ideal gas equation

$$PV = nRT$$

or in molecular form,

$$PV = Nk_BT$$

where N is number of molecules and k_B is Boltzmann constant.

Kinetic interpretation

Gas pressure arises from molecular collisions with the container walls. Temperature is proportional to average translational kinetic energy.

Mean translational energy

For an ideal monatomic gas,

$$U = \frac{3}{2}Nk_BT$$

and molar internal energy is

$$U = \frac{3}{2}nRT$$

11. Specific heats of gases

For ideal gases:

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

$$C_P = \left(\frac{\partial H}{\partial T}\right)_P$$

where H is enthalpy.

Mayer's relation

$$C_P - C_V = R$$

for one mole of an ideal gas.

Monatomic ideal gas

$$C_V = \frac{3}{2}R, C_P = \frac{5}{2}R$$

12. Enthalpy

Enthalpy is defined as

$$H = U + PV$$

It is especially useful for processes at constant pressure.

Differential form

$$dH = dU + PdV + VdP$$

At constant pressure,

$$q_p = \Delta H$$

13. The second law

The second law introduces the direction of natural processes. It explains why heat flows spontaneously from hot bodies to cold bodies and why no engine can be perfectly efficient.

Kelvin-Planck statement

No engine operating in a cycle can convert all absorbed heat into work.

Clausius statement

Heat cannot spontaneously flow from a colder body to a hotter body without external work.

Entropy statement

For any spontaneous process, the entropy of the universe increases.

14. Entropy

Entropy is a thermodynamic state function that measures energy dispersal and the degree of irreversibility.

Definition for reversible process

$$dS = \frac{\delta Q_{\mathrm{rev}}}{T}$$

Important ideas

- Entropy increases in irreversible processes.
- Phase changes typically involve entropy change.
- For a perfect crystal at absolute zero, entropy approaches zero.

15. Third law

The third law states that the entropy of a perfect crystal approaches zero as temperature approaches absolute zero.

This law provides an absolute reference point for entropy and explains why absolute zero cannot be reached in finite steps.

16. Thermodynamic potentials

Thermodynamic potentials are useful for predicting spontaneity under different constraints.

Helmholtz free energy

$$F = U - TS$$

Useful at constant temperature and volume.

Gibbs free energy

$$G = H - TS$$

Useful at constant temperature and pressure.

Spontaneity criterion

At constant $T$and P,

$$\Delta G < 0$$

for a spontaneous process.

17. Heat engines and refrigerators

A heat engine converts thermal energy into work by operating between two reservoirs. A refrigerator does the reverse by using work to transfer heat from cold to hot.

Efficiency of engine

$$\eta = \frac{W}{Q_H} = 1 - \frac{Q_C}{Q_H}$$

Carnot engine

The Carnot engine is an ideal reversible engine and gives the maximum possible efficiency between two temperatures.

Carnot efficiency

$$\eta_C = 1 - \frac{T_C}{T_H}$$

18. Cyclic processes and PV diagrams

In a cyclic process, the system returns to its initial state, so $\Delta U = 0$. The work done equals the area enclosed by the cycle on a PV diagram.

Common processes

- Isothermal process: temperature constant.
- Adiabatic process: no heat exchange.
- Isochoric process: volume constant.
- Isobaric process: pressure constant.

Adiabatic condition

For a reversible adiabatic process of an ideal gas,

$$PV^{\gamma} = \text{constant}$$

where $\gamma = C_P / C_V$.

19. Heat transfer

Thermal physics also studies conduction, convection, and radiation as modes of heat transfer.

Conduction

$$\frac{dQ}{dt} = kA\frac{\Delta T}{L}$$

where k is thermal conductivity.

Radiation

$$P = \epsilon\sigma A T^4$$

from the Stefan-Boltzmann law for thermal radiation.

Newton's law of cooling

$$\frac{dT}{dt} \propto (T - T_s)$$

where T_s is surrounding temperature.

20. Phase change and latent heat

During melting, boiling, and sublimation, temperature may remain constant while heat is absorbed or released.

Latent heat formula

$$Q = mL$$

where $L$is latent heat.

Types

- Latent heat of fusion.
- Latent heat of vaporization.

21. Maxwell relations and advanced ideas

In advanced thermodynamics, the differential forms of U, H, F, and Glead to Maxwell relations. These connect measurable partial derivatives and are central to theoretical physics.

Example differential forms

$$dU = TdS - PdV$$

$$dH = TdS + VdP$$

$$dF = -SdT - PdV$$

$$dG = -SdT + VdP$$

22. Important formula list

- $T_K = T_{^\circ C} + 273.15$.
- $PV = nRT$.
- $dW = -P\,dV$.
- $\Delta U = Q + W$.
- $Q = mc\Delta T$.
- $Q = mL$.
- $H = U + PV$.
- $dS = \delta Q_{\text{rev}}/T$.
- $G = H - TS$.
- $\eta_C = 1 - T_C/T_H$.
- $PV^\gamma =$ constant.
- $\Delta G < 0$for spontaneity at constant T, P.

23. Exam-oriented distinctions

Heat vs temperature

- Heat is energy in transit.
- Temperature is a measure of thermal state.

Internal energy vs enthalpy

- Internal energy includes all microscopic energy.
- Enthalpy is useful for constant-pressure processes.

Reversible vs irreversible

- Reversible processes are idealized and quasi-static.
- Irreversible processes occur naturally and increase entropy.

24. Summary

Thermal physics and thermodynamics provide the foundation for understanding heat, temperature, work, energy conservation, entropy, and equilibrium. In BSc Physics, the subject is essential because it connects laboratory observations with the deep principles that govern engines, gases, phase changes, and spontaneous processes.